

ME 142
ENGINEERING DRAWING
&
GRAPHICS
(Lettering)

ABCDEFGHIJKLMNOPQRSTUVWXYZ

UVWXYZABCDEFGHIJKLM

NOPQRSTUVWXYZABCDEF

Text on Drawings

Text on engineering drawing is used :

- To communicate nongraphic information.
- As a substitute for graphic information, in those instance where text can communicate the needed information more clearly and quickly.

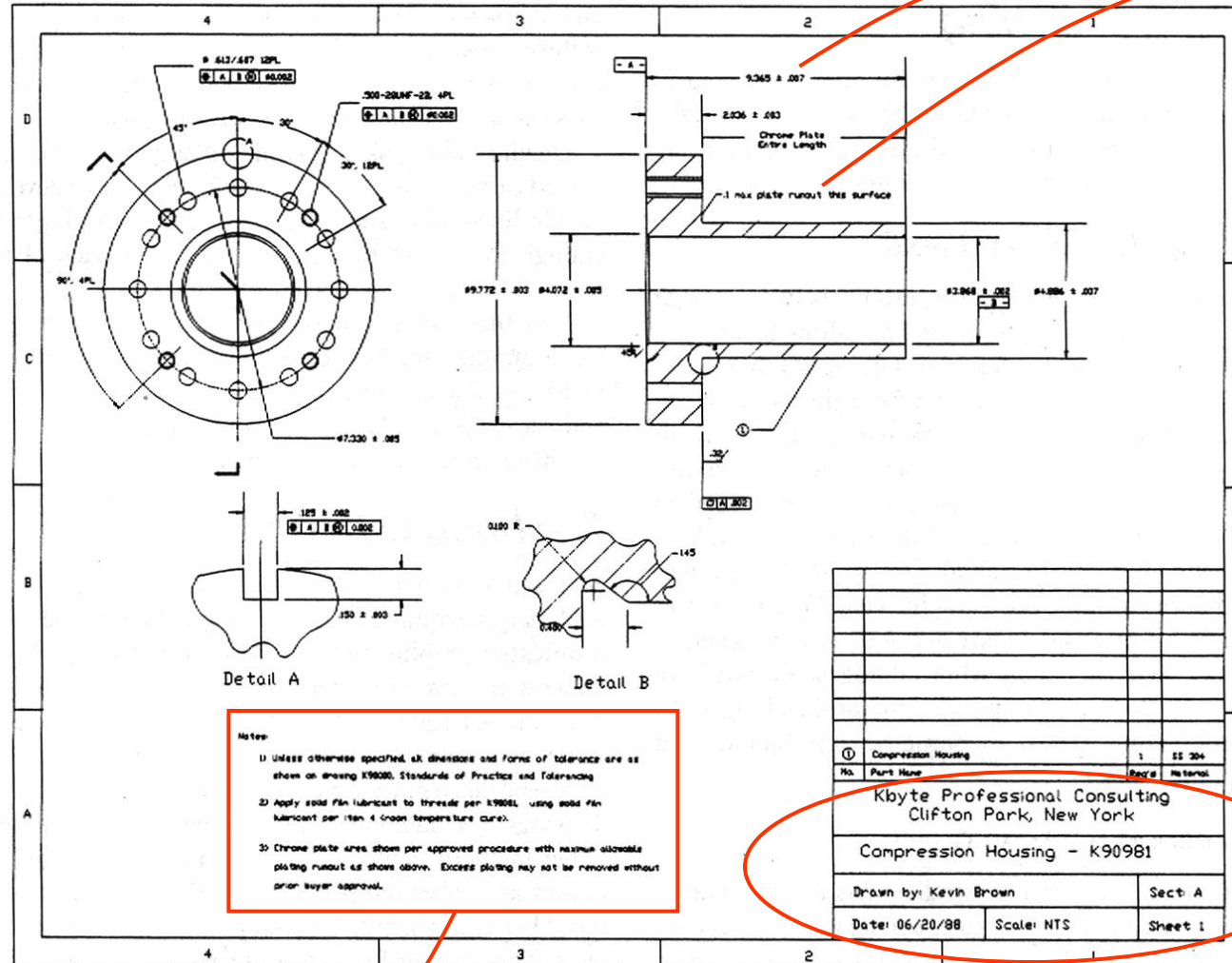
Thus, it must be written with

Legibility - shape
 - space between letters and words

Uniformity - size
 - line thickness

Example Placement of the text on drawing

Dimension & Notes



Notes

Title Block

Lettering Standard

ANSI Standard

- Use a Gothic text style, either inclined or vertical.
- Use all capital letters.
- Use 3 mm for most text height.
- Space between lines of text is **at least** 1/3 of text height.

This course

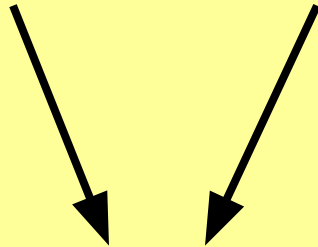
- Use only a vertical Gothic text style.
- Use both capital and lower-case letters.
- Same. For letters in title block it is recommend to use 5~8 mm text height
- N/A.
Follows ANSI rule.

Basic Strokes

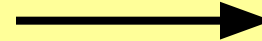
Straight



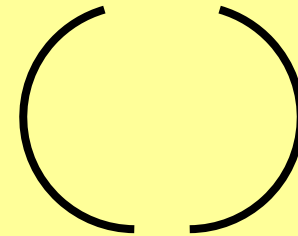
Slanted



Horizontal



Curved

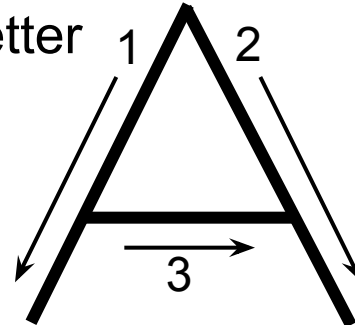


Examples : *Application of basic stroke*

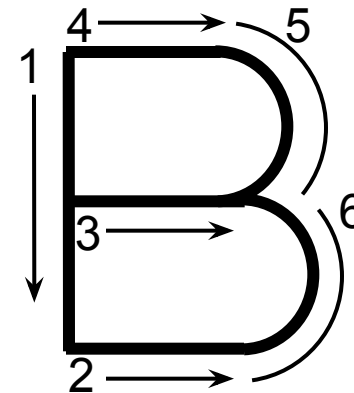
“I” letter



“A” letter

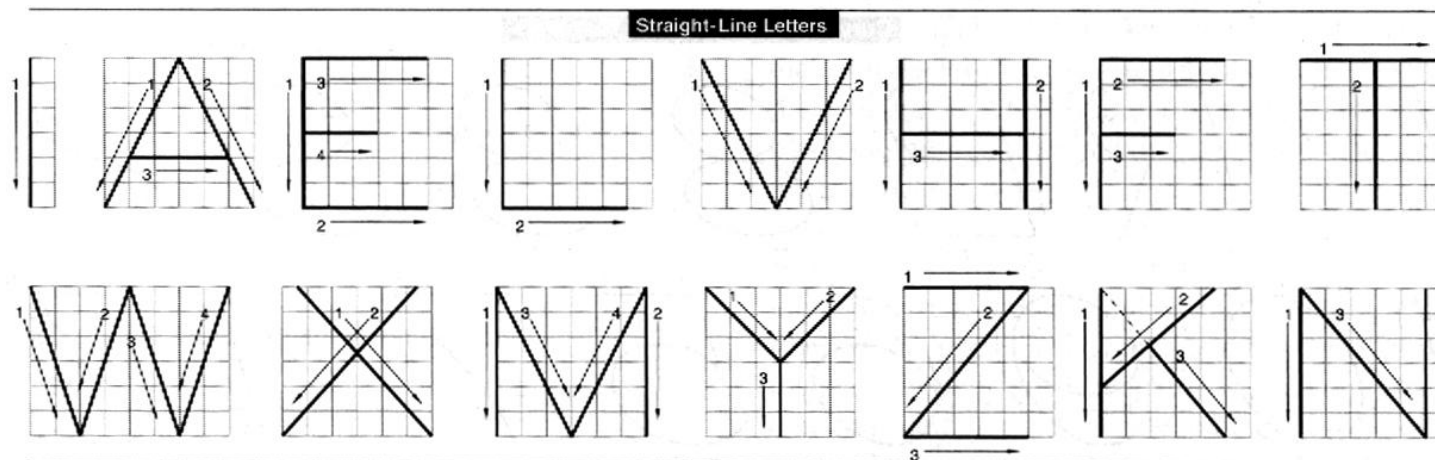


“B” letter

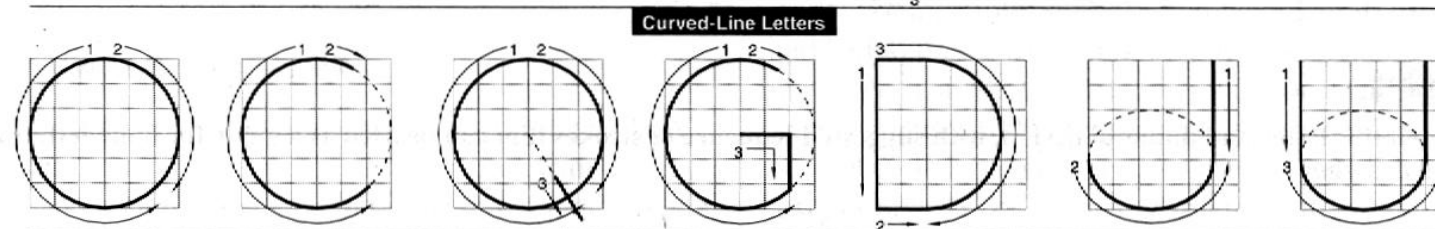


Upper-case letters & Numerals

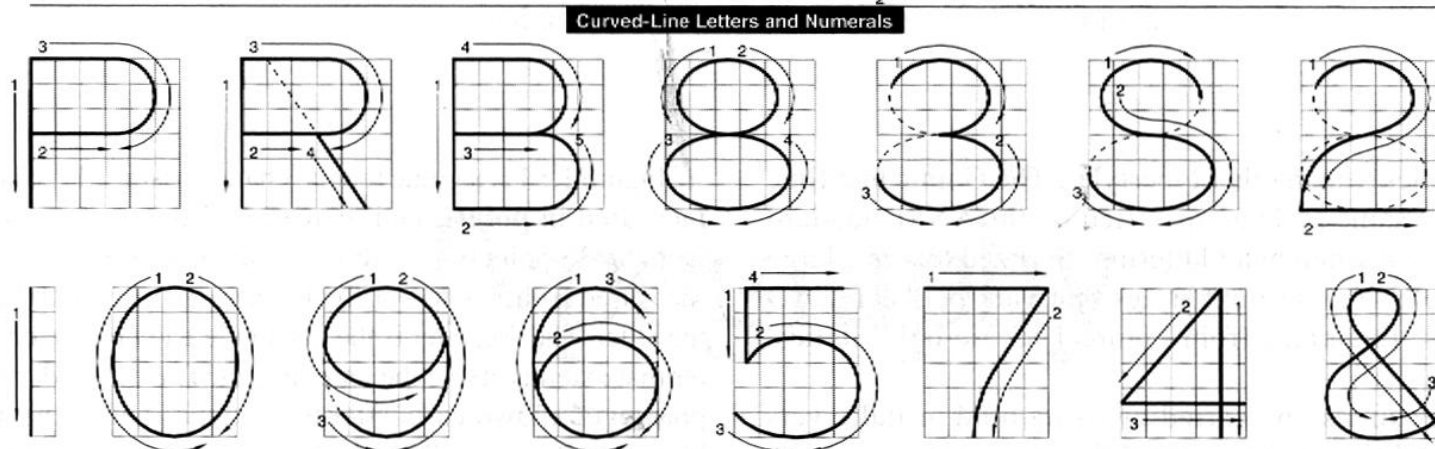
Straight line
letters



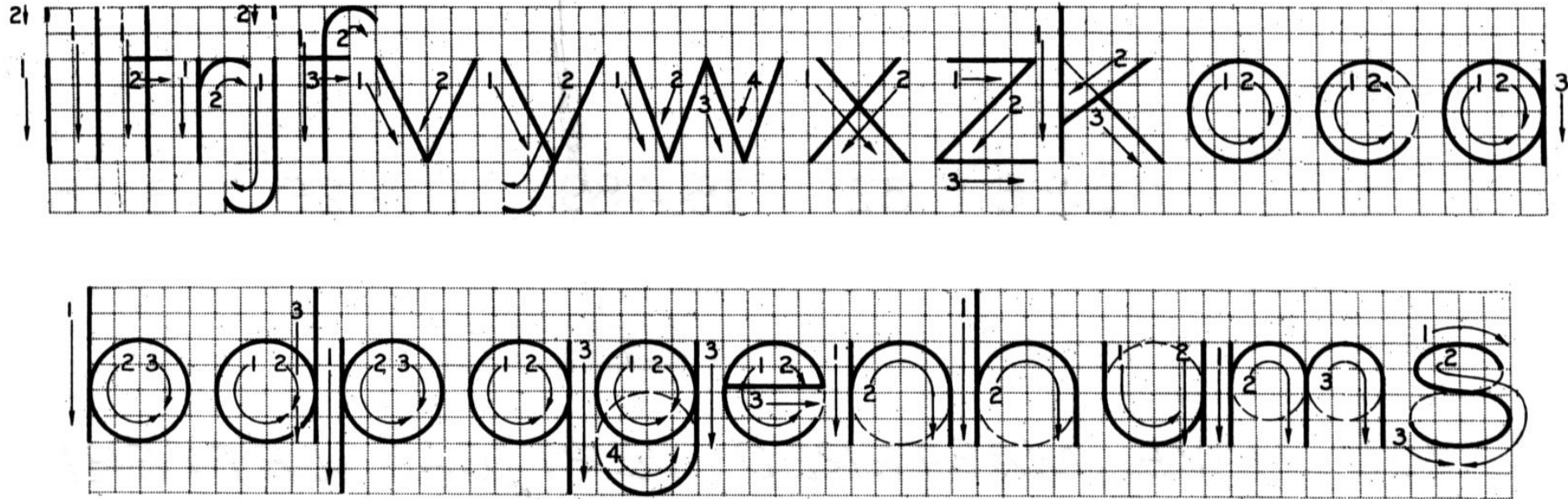
Curved line
letters



Curved line
letters &
Numerals



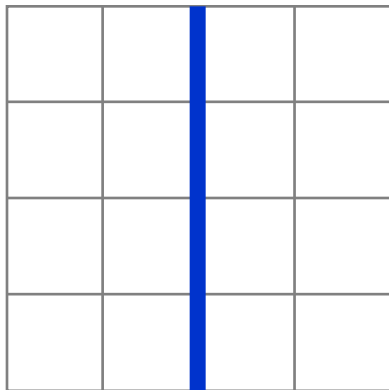
Lower-case letters



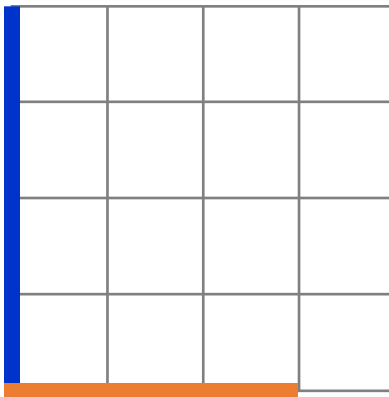
- The text's body height is about $\frac{2}{3}$ the height of a capital letter.

Stroke Sequence

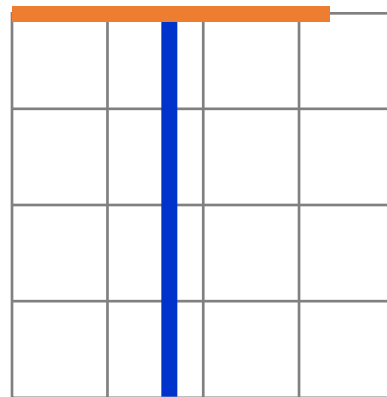
I



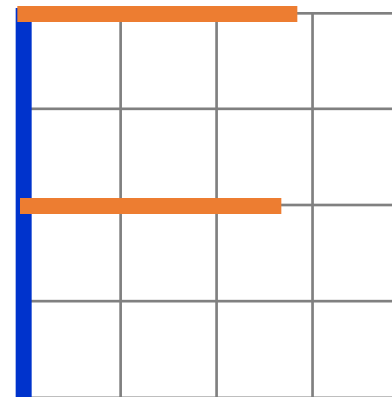
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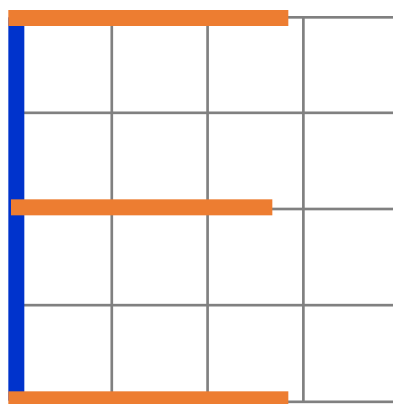
T



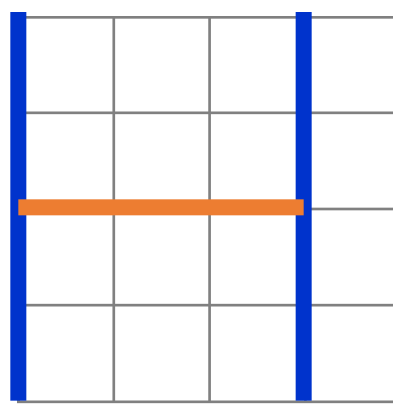
F



E

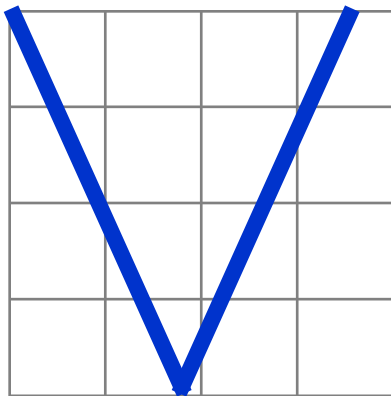


H

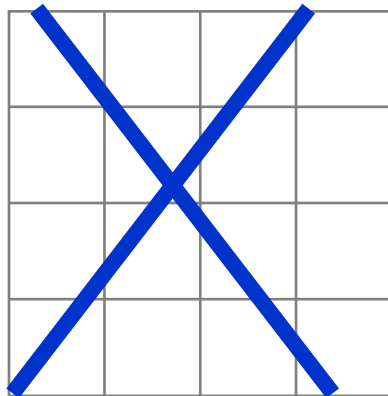


Stroke Sequence

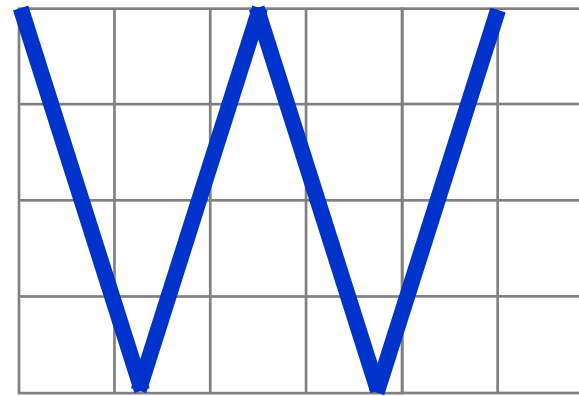
V



X

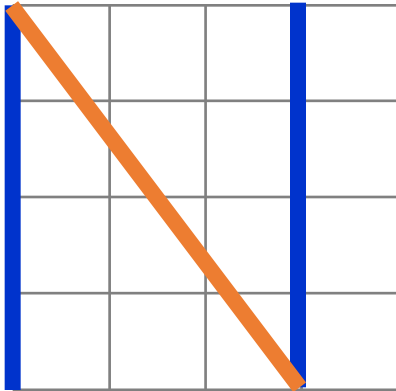


W

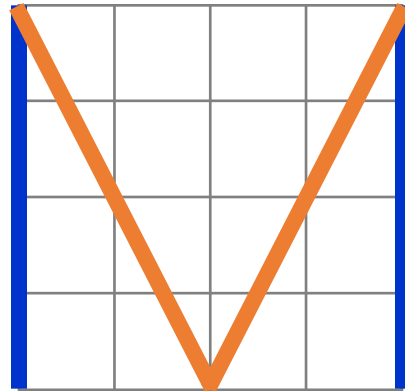


Stroke Sequence

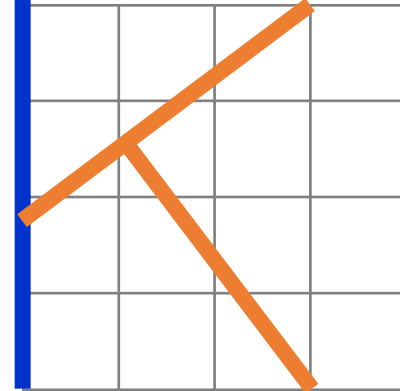
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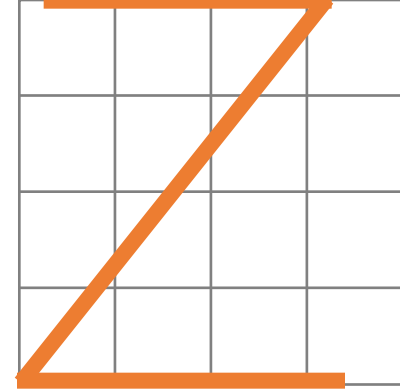
M



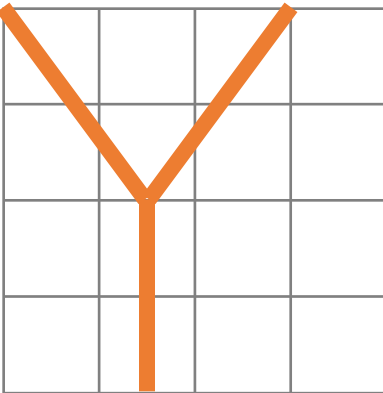
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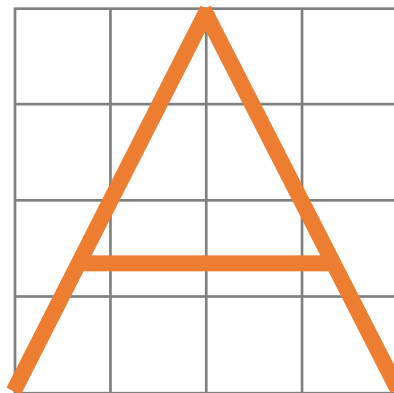
Z



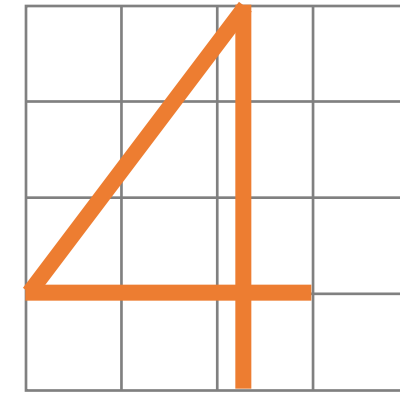
Y



A

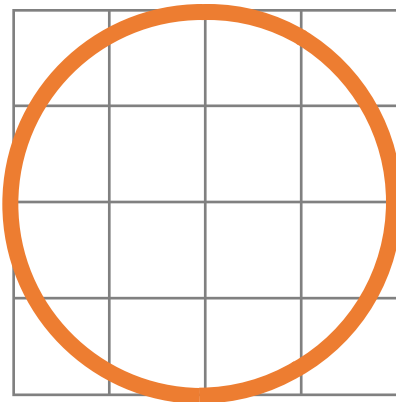


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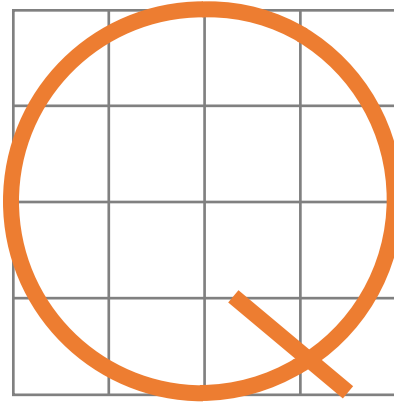


Stroke Sequence

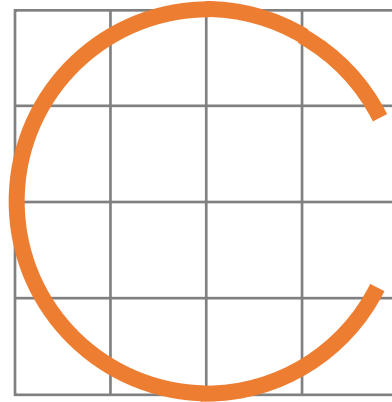
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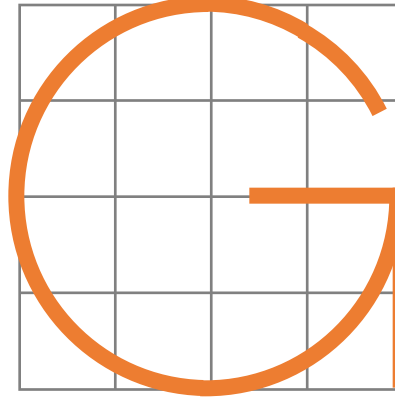
Q



C

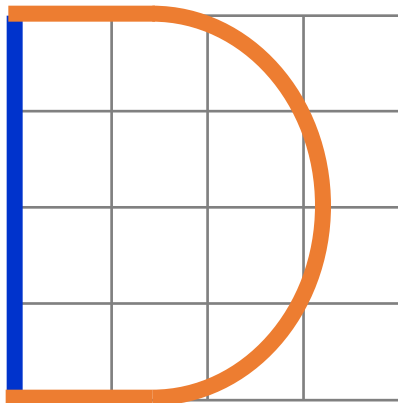


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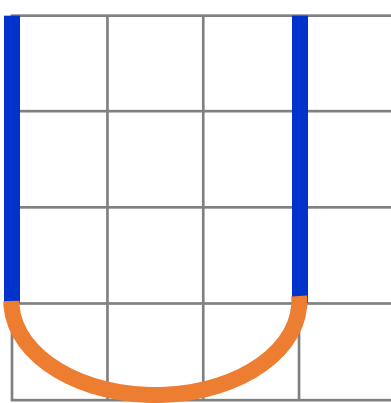


Stroke Sequence

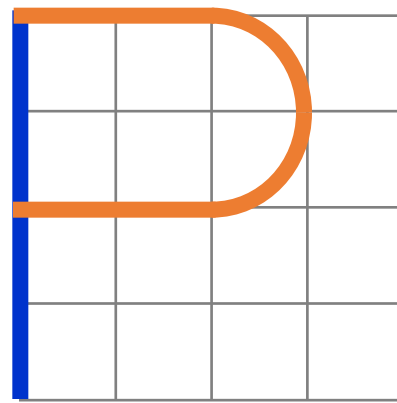
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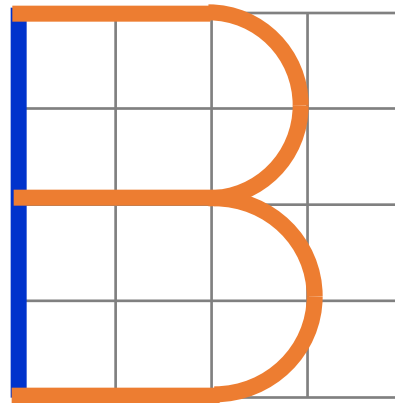
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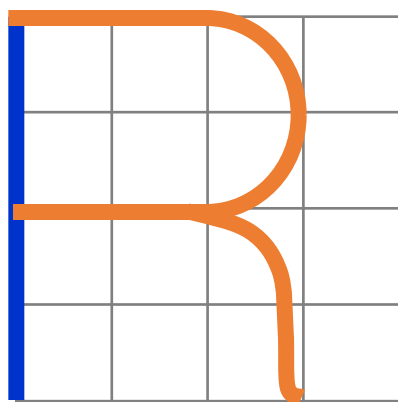
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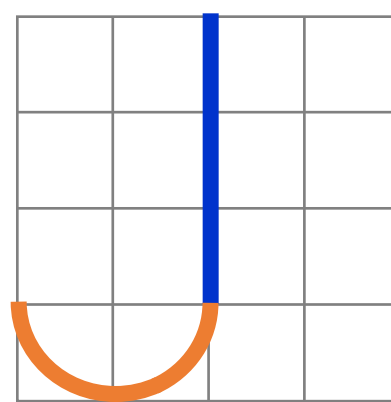
B



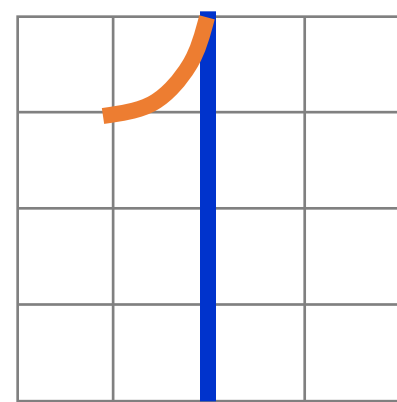
R



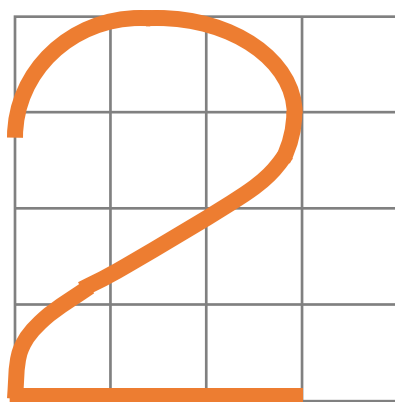
J



1

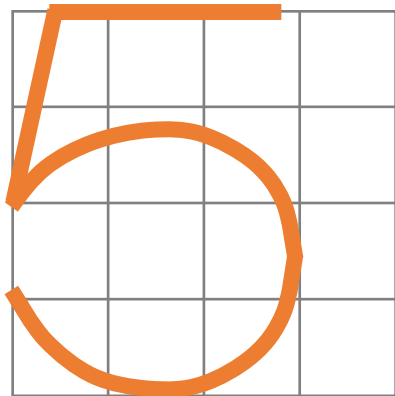


2

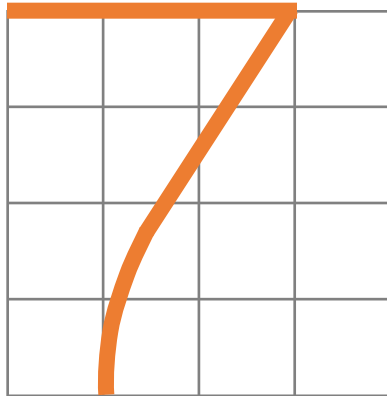


Stroke Sequence

5

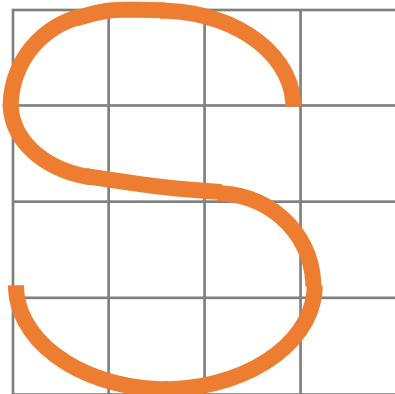


7

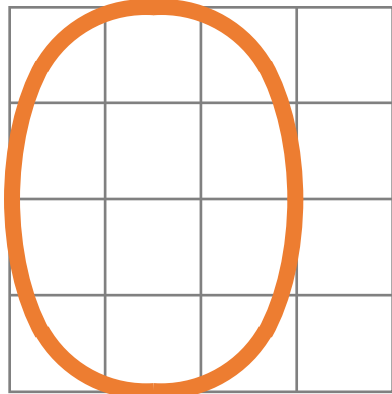


Stroke Sequence

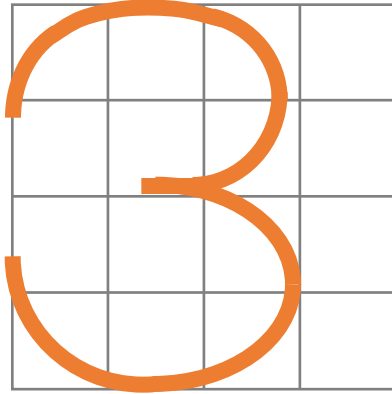
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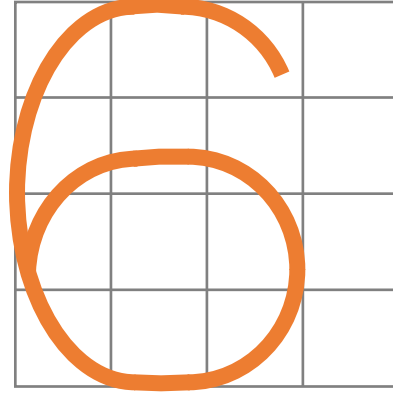
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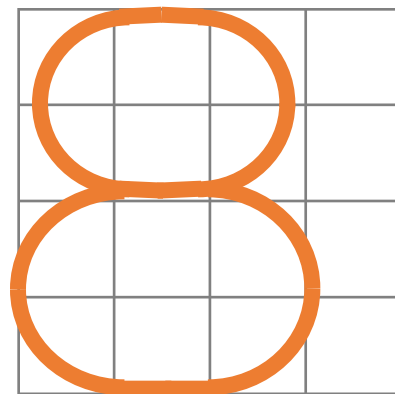
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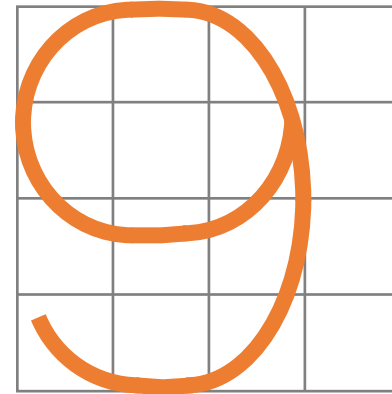
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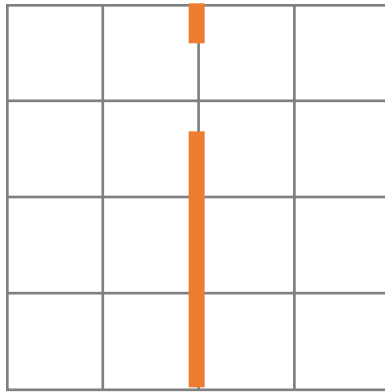
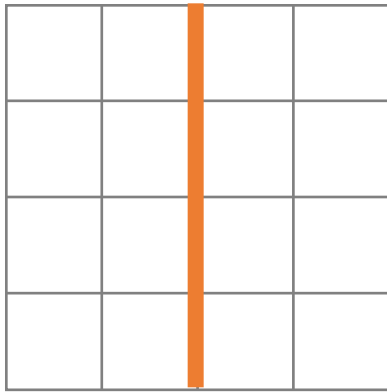
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9

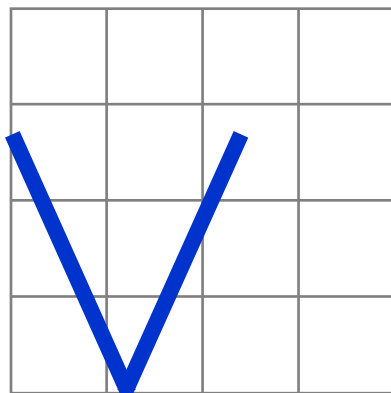


Stroke Sequence

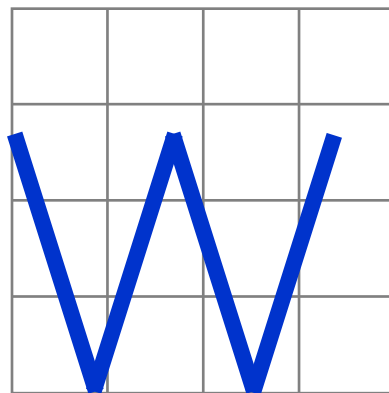


Stroke Sequence

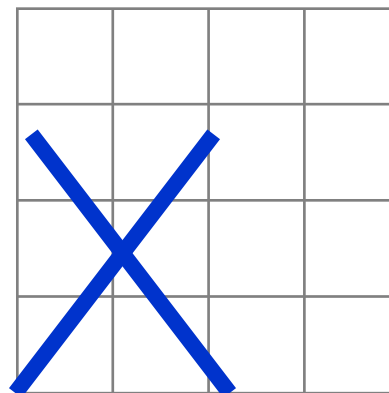
v



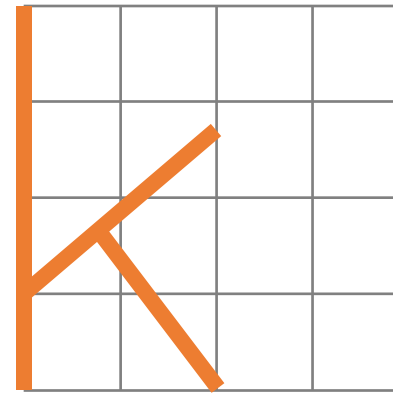
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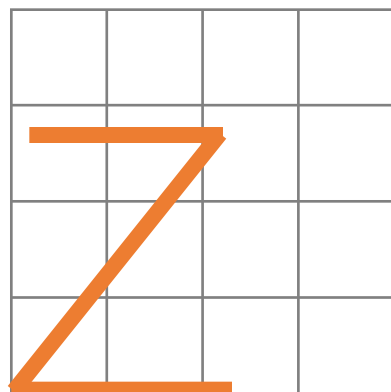
x



k

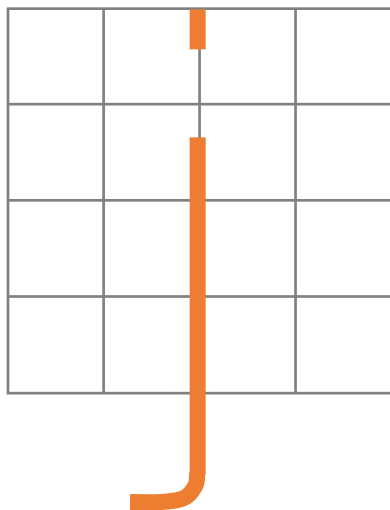


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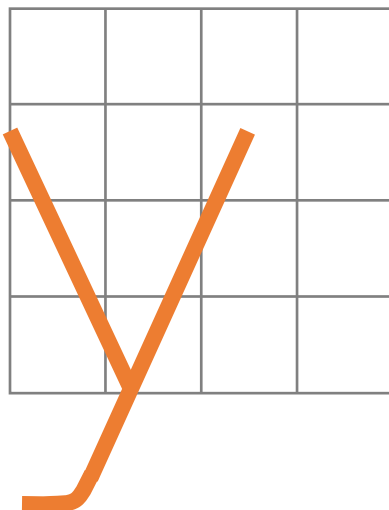


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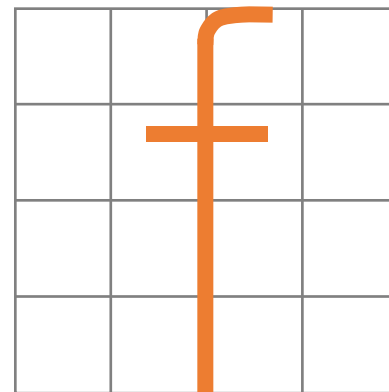
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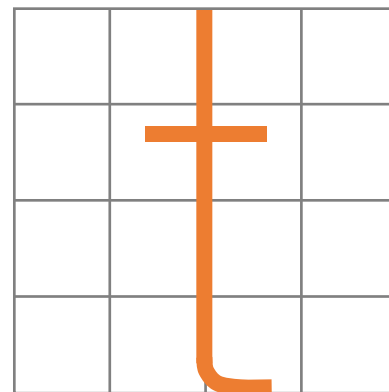
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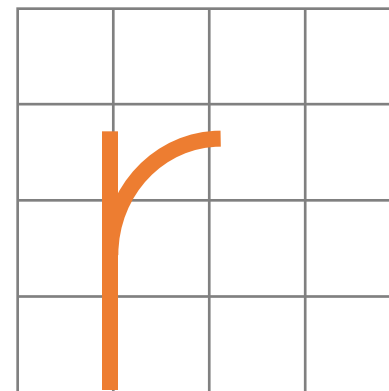
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t

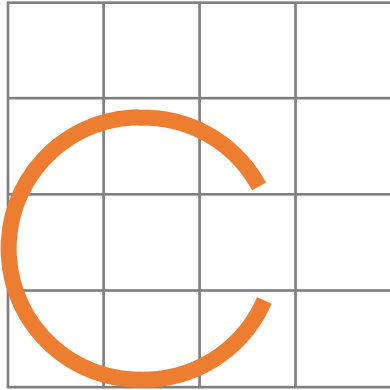


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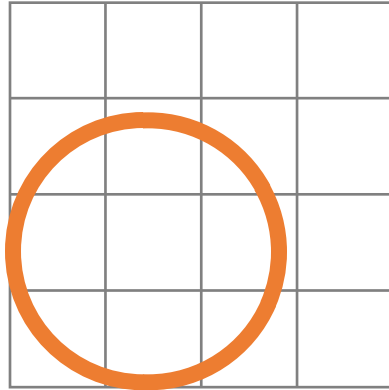


Stroke Sequence

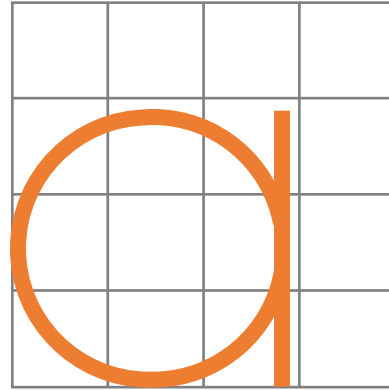
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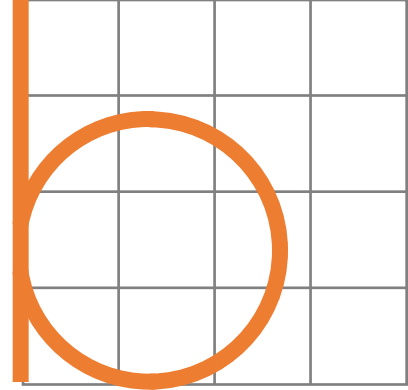
o



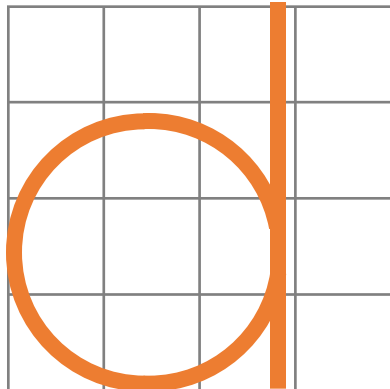
a



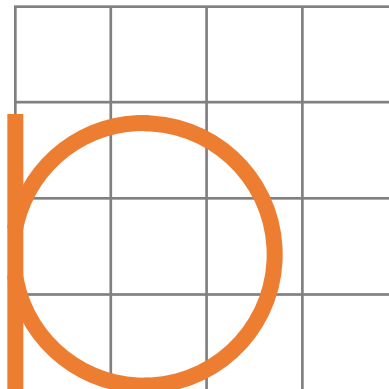
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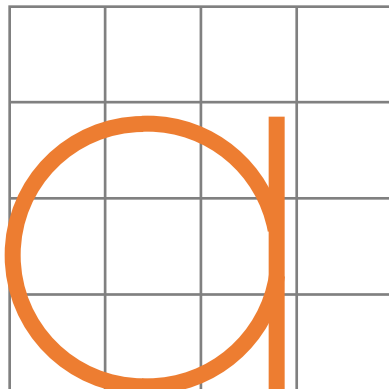
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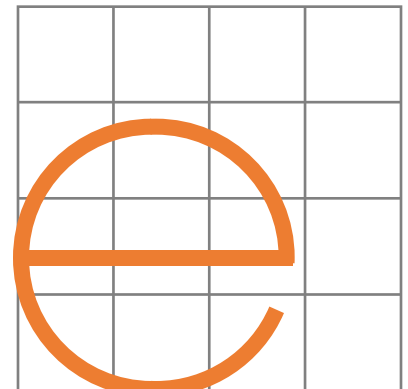
p



q

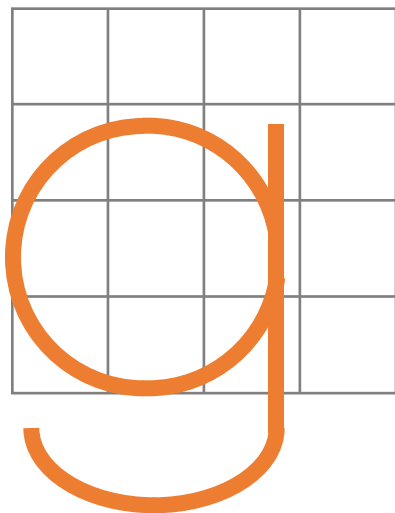


e

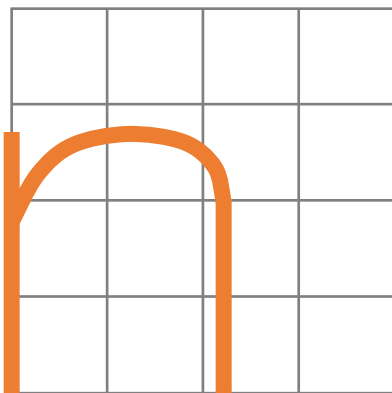


Stroke Sequence

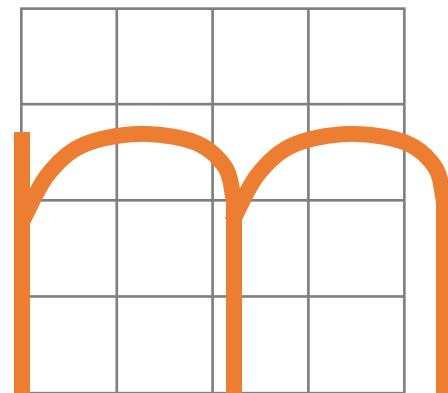
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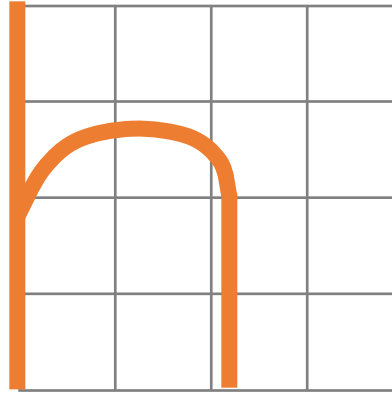
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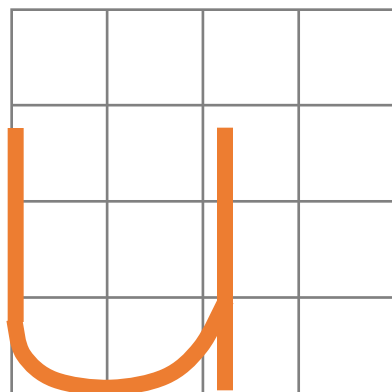
m



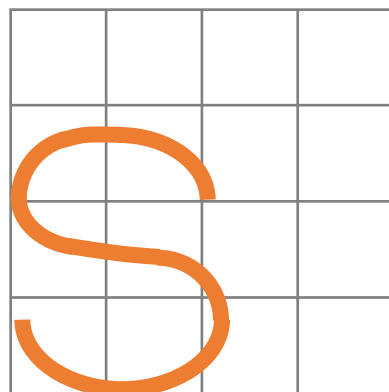
h



u



s



Word Composition

Look at the same word having different spacing between letters.

A) Non-uniform spacing

JIRAPONG



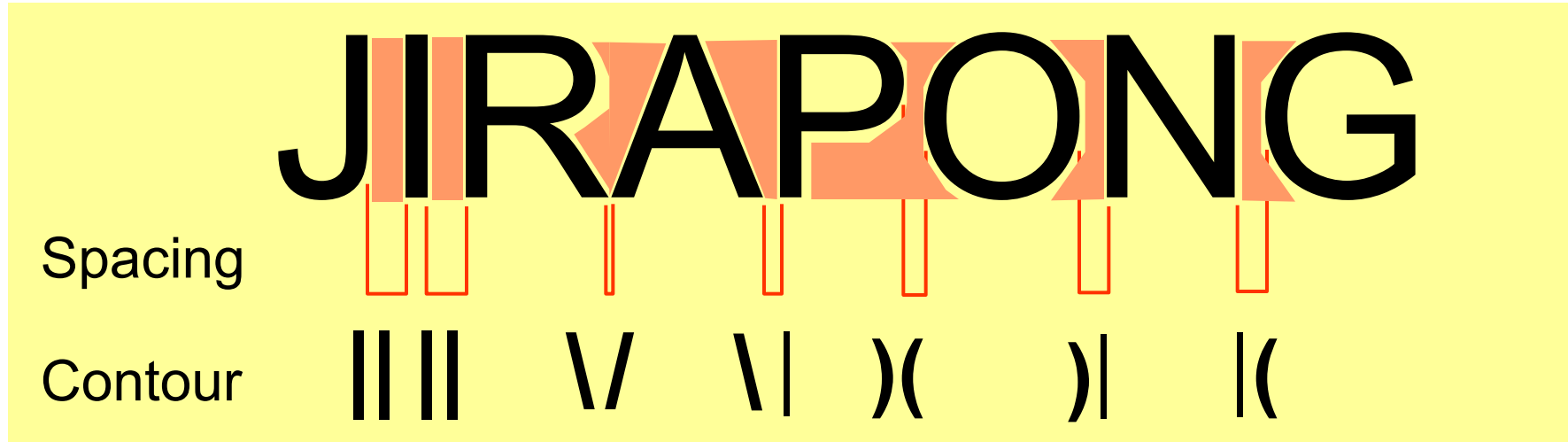
B) Uniform spacing

JIRAPONG



Which one is easier to read ?

Word Composition



General conclusions are:

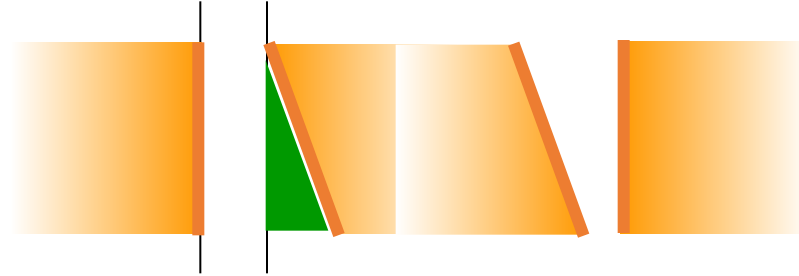
- Space between the letters depends on the contour of the letters at an adjacent side.
- Good spacing creates approximately equal ***background area*** between letters.

Space between Letters

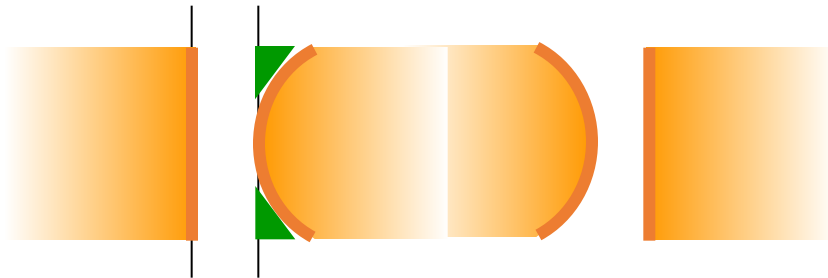
1. Straight - Straight



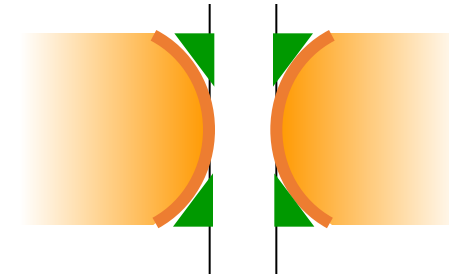
3. Straight - Slant



2. Straight - Curve



4. Curve - Curve



Space between Letters

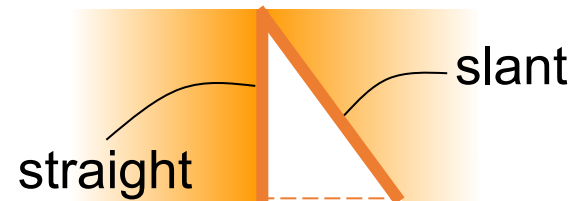
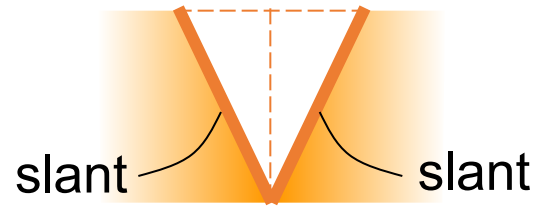
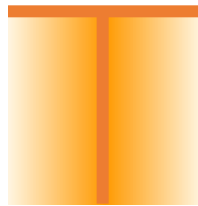
5. Curve - Slant



6. Slant - Slant



7. The letter “L” and “T”



Example : Good and Poor Lettering

ESTIMATE

GOOD

EstiMaTE

Not uniform in style.

ESTIMATE

Not uniform in height.

ESTIMATE

ESTIMATE

Not uniformly vertical or inclined.

ESTIMATE

ESTIMATE

Not uniform in thickness of stroke.

ESTIMATE

Area between letters not uniform.

ESTIMATE

Area between words not uniform.

ABILITY WILL NEVER CATCH UP
WITH THE DEMAND FOR IT

Sentence Composition

- Leave the space between words equal to the space requires for writing a letter “O”.

Example

ALL O DIMENSIONS O ARE O IN
MILLIMETERS O UNLESS
OTHERWISE O SPECIFIED.